

HAT-P-54b

R-1790

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_170204 · created 2026-08-03T00:39:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457788.84553 +/- 0.00033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.045
Transit depth (Rp/Rs)^2	2.77 +/- 1.49 [%]
Orbital Inclination (inc)	89.79 +/- 1.88
Airmass coefficient 1 (a1)	0.911 +/- 0.024
Airmass coefficient 2 (a2)	0.046 +/- 0.016
Scatter in the residuals of the lightcurve fit is	2.69 %
Best Comparison Star	#1 - [477, 378]
Optimal Aperture	2.49
Optimal Annulus	5.12
Transit Duration (day)	0.0711 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.045 vs 0.1572 (deviation 0.17σ)
Duration fit vs published	0.0711 d vs 0.075 d (deviation 1.05σ)
Mid-time O–C	-2.6 min
Depth SNR	2.9
Residual scatter	2.69 %

Light curve

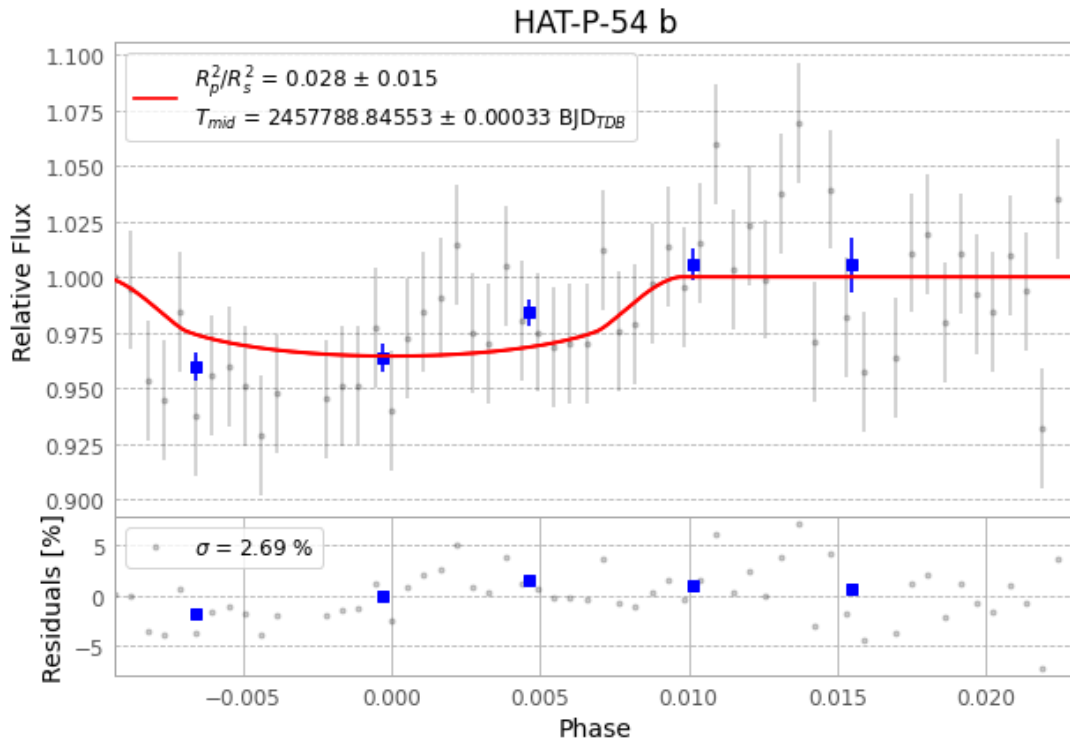


Figure 1 — FinalLightCurve_HAT-P-54 b_2017-02-04.png (SHA-256 ea06428db57c0f1a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [336, 294], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7810c7a7282ae3d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), HATP-54170204071814.FITS → HATP-54170204101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-170204.log (8fe1fd5d047d...), FinalLightCurve_HAT-P-54 b_2017-02-04.png (ea06428db57c...), FinalLightCurve_HAT-P-54 b_2017-02-04.csv (def2ab2c4245...)

Compute resources

Wall time 360.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 00:40Z.